

A Certified Computation of the Cyclic Ramsey Number $R_{\text{cyc}}(K_4, P_6^{\text{alt}})$

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Abstract

We prove the exact cyclic Ramsey number

$$R_{\text{cyc}}(K_4, P_6^{\text{alt}}) = 16.$$

The only computer-assisted step classifies the extremal colorings on 15 vertices. A SAT instance asserting the existence of an induced blue three-vertex path in a coloring with no red K_4 and no blue cyclic P_6^{alt} is unsatisfiable. Its ASCII DRAT certificate was checked independently by `drat-trim`. It follows that every extremal blue graph is $K_5 \dot{\cup} K_5 \dot{\cup} K_5$. A short deletion argument then proves the upper bound on 16 vertices. We provide independent encoders, validators, hashes, and a complete proof certificate.

1 Introduction

Bašić, Damnjanović, Stevanović, and Stošić introduced and systematically studied cyclic Ramsey numbers for ordered graphs [1]. Their Conjecture 4.23 predicts

$$R_{\text{cyc}}(K_a, P_b^{\text{alt}}) = 1 + (a - 1)(b - 1).$$

The case $(a, b) = (4, 6)$ therefore predicts the value 16. The lower bound is given by the familiar coloring whose blue graph is three disjoint copies of K_5 . The purpose of this note is to certify the matching upper bound.

The alternating path P_6^{alt} has vertices $0, 1, \dots, 5$ and, in path order, is

$$0, 5, 1, 4, 2, 3.$$

A cyclic copy preserves the orientation of the cyclic vertex order: cyclic rotations are allowed, while reflections are not. All computations below use this convention.

2 Reduction to a finite classification

We regard the blue edges as a graph B ; nonedges of B are red.

Lemma 1. *If a graph has no induced P_3 , then each of its connected components is a clique.*

Proof. If a connected component were not complete, choose two nonadjacent vertices at minimum distance. The first three vertices of a shortest path between them induce a P_3 . $\square$

Proposition 2 (Certified classification). *Let the vertices $0, \dots, 14$ carry their natural cyclic order. If a red-blue coloring has neither a red K_4 nor a blue cyclic P_6^{alt} , then its blue graph is*

$$K_5 \dot{\cup} K_5 \dot{\cup} K_5.$$

Proof. The computer-assisted part proves that the blue graph has no induced P_3 ; the encoding and certificate are described in Section 4. By Lemma 1, the blue graph is a disjoint union of cliques. There are at most three components, since one vertex from each of four components would form a red K_4 . Every component has order at most five, since a blue K_6 contains a cyclic P_6^{alt} in every cyclic ordering. Fifteen vertices distributed among at most three components of order at most five must therefore form exactly three copies of K_5 . $\square$

3 The Ramsey number

Theorem 3.

$$R_{\text{cyc}}(K_4, P_6^{\text{alt}}) = 16.$$

Proof. For the lower bound, color the edges within each of three five-vertex parts blue and all edges between parts red. A blue connected subgraph has at most five vertices, and a red clique has at most one vertex in each part. Thus this 15-vertex coloring contains neither a blue cyclic P_6^{alt} nor a red K_4 .

For the upper bound, suppose that a valid coloring on 16 cyclically ordered vertices exists and delete any vertex x . Proposition 2 partitions the remaining vertices into three blue cliques A, B, C , each of order five, with every edge between distinct parts red. If x had a red neighbor $a \in A$, $b \in B$, and $c \in C$, then $\{x, a, b, c\}$ would be a red K_4 . Hence x is blue-complete to at least one part. That part together with x is a blue K_6 , which contains a cyclic P_6^{alt} . This contradiction proves the upper bound. $\square$

4 Encoding and certification

For each unordered edge ij of K_{15} , let x_{ij} be true exactly when ij is blue. There are 105 edge variables. For every four-set we add the clause

$$\bigvee_{ij \in E(K_4)} x_{ij},$$

which excludes an all-red K_4 . For every orientation-preserving cyclic embedding of P_6^{alt} , we add

$$\bigvee_{ij \in E(P_6^{\text{alt}})} \neg x_{ij},$$

which excludes an all-blue copy. Reflected embeddings are not added.

If a non-cluster blue graph exists, it contains an induced blue P_3 . Its unique center may be moved to vertex 0 by a cyclic rotation. Its endpoints are an unordered pair $\{u, w\} \subseteq \{1, \dots, 14\}$, giving 91 cases. For each pair introduce a selector s_{uw} . Exactly one selector is true, and the implication clauses

$$s_{uw} \implies (x_{0u} \wedge x_{0w} \wedge \neg x_{uw})$$

force its induced blue P_3 . Requiring exactly one selector merely chooses a witness; it does not assert that the graph has only one induced P_3 .

The resulting DIMACS instance has 196 variables and 39,859 clauses: 35,490 clauses from the base encoding, 273 selector implications, one at-least-one selector clause, and 4,095 pairwise at-most-one clauses. Kissat reports the instance UNSAT and emits an ASCII DRAT proof. The proof was checked with an independently built `drat-trim`; the checker reports `s VERIFIED`, with 19,912 input clauses in the core, 3,427,067 proof lemmas in the core, and 123,107,659 resolution steps.

As an encoder check, we generated the base formula independently from the definitions and compared it with the unmodified generator from the authors' supplementary repository [2].

After canonicalization, both the clause sets and duplicate multiplicities agree. Both encoders produce 105 variables, 35,490 raw clauses, and 16,380 distinct clauses. We also solved the 91 selected cases individually; every case was UNSAT. This second run is a diagnostic cross-check, while the single DRAT certificate is the formal computational evidence for Proposition 2.

5 Reproducibility

The proof package contains the independent encoder, the combined-CNF reconstructor, model validators, unit tests, SHA-256 hashes, solver metadata, the DIMACS instance, the ASCII DRAT proof, and the complete checker log. The principal hashes are

CNF: f59fda6c63246901b7d43f35bcca4adcbb61efff0f09cfce9b44d5b1cdd9da03,
 DRAT: 38a4f3ffabca676a8e1f77d090db8fadaa92d8328ee4061bf5cc55fbfc877d78.

The verification script first audits the encodings and hashes, and can then rerun `drat-trim` on the 817,727,304-byte proof. Software versions and all remaining artifact hashes are recorded in the machine-readable manifest.

AI contribution and provenance statement

The initial structural direction—greedy partitioning and deletion of middle labels—was proposed by Julien Menet. GPT-5.6-SOL and Qwen3.8max subsequently made substantial intellectual and technical contributions, including development and refinement of the structural reductions, formulation of the finite classification problem, SAT-encoding design, software implementation, solver and certificate engineering, computational experiments, adversarial verification, and initial manuscript drafting. Menet directed the project, selected and evaluated research directions, reviewed the mathematical arguments and computational evidence, approved the final manuscript, and accepts responsibility for all claims and released artifacts. The AI systems are credited as non-author research contributors because current scholarly authorship conventions require a human author capable of assuming responsibility for the work.

References

- [1] N. Bašić, I. Damnjanović, D. Stevanović, and I. Stošić, *Some results on small ordered and cyclic Ramsey numbers*, arXiv:2604.16188 (2026), <https://arxiv.org/abs/2604.16188>.
- [2] I. Damnjanović, *Supplementary code for ordered and cyclic Ramsey numbers*, GitHub repository, commit 8ce4e0ab01f1321cb9555703cd8def4d7a1c8511, <https://github.com/Ivan-Damnjanovic/ord-ram-num>.